Data Science Jobs: Sentiments, Salaries, and Direction on Social Media.

Word count: 1494

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Abstract

The era of AI technologies has led to diverse sentiments among people regarding data science jobs, prompting an exploration of these sentiments and their correlation with salary. The purpose of this project is to analyze the relationship between sentiments expressed about data science domain jobs and their corresponding salaries. For this analysis, two different data sources were used, comprising a total of 53,341 rows of data: 36,847 posts from Reddit and 16,494 rows of data related to data science job salaries. The overall results indicate that data science job professionals take a balanced perspective on these jobs, and salary is weekly correlated with their sentiments expressed in Reddit posts. The findings show variability in average salaries among different jobs, with the term hiring being more prominent in discussions within the field. This study provides insights into the sentiments and discussions of data science professionals, offering a balanced view of the job market and opportunities within the field.

1. Research Design

This study adopts an inductive research design, aiming to explore the relationship between data science job salaries and sentiments expressed on Reddit. The research questions are: What is the correlation between different data science job's salary and sentiments in data science job discussions on Reddit? and why do these sentiments vary across different job titles within the data science domain? The process model starts with data collection from Kaggle and Reddit, followed by sentiment analysis, correlation analysis, and topic modeling. This model enables for development and deeper insights as new patterns emerge from the data, ensuring a comprehensive understanding of the relationship between jobs and sentiments on Reddit.

2. Data Collection

For this study, identified two main datasets: one from Kaggle and the other from Reddit. The Kaggle dataset includes a wide range of data science jobs and corresponding salaries from different countries and four different years, which is why chose the Kaggle (salaries) dataset as the first dataset. The second main dataset is from Reddit, initially scraped from subreddits relevant to data science jobs such as 'datascience', 'MachineLearning', and 'cscareerquestions'. However, this data scraping was not productive because approximately 3000 posts were scraped, and consolidating the posts into one data frame was tedious. Therefore, fetching posts by keywords was found to be useful, as it helped scrape the most relevant Reddit posts for this study. The keywords were identified from Kaggle's dataset's unique job titles, with 152 unique job titles extracted from Kaggle's 16,494 rows of data. Using these job titles as keywords, 36,847 posts were scraped by enumerating the keywords, with the for loop and exiting using a break statement. Additionally, exceptions were handled using the prawcore.exceptions library, and try and except blocks were used in the loops. The Kaggle dataset was called df, and for empty cells and NaNs, the isna() method was used for df cleaning. For better future analysis, a new column for the job nature corresponding to the job title was created using a dictionary, and the job nature was rewritten to a new column in the unique_values_title_column CSV file. While cleaning the Reddit dataset, I filtered out any keywords that did not scrape posts from Reddit and dropped the URL column because its unnecessary. Finally, merged the Reddit dataset with the unique_values_title_column dataset.

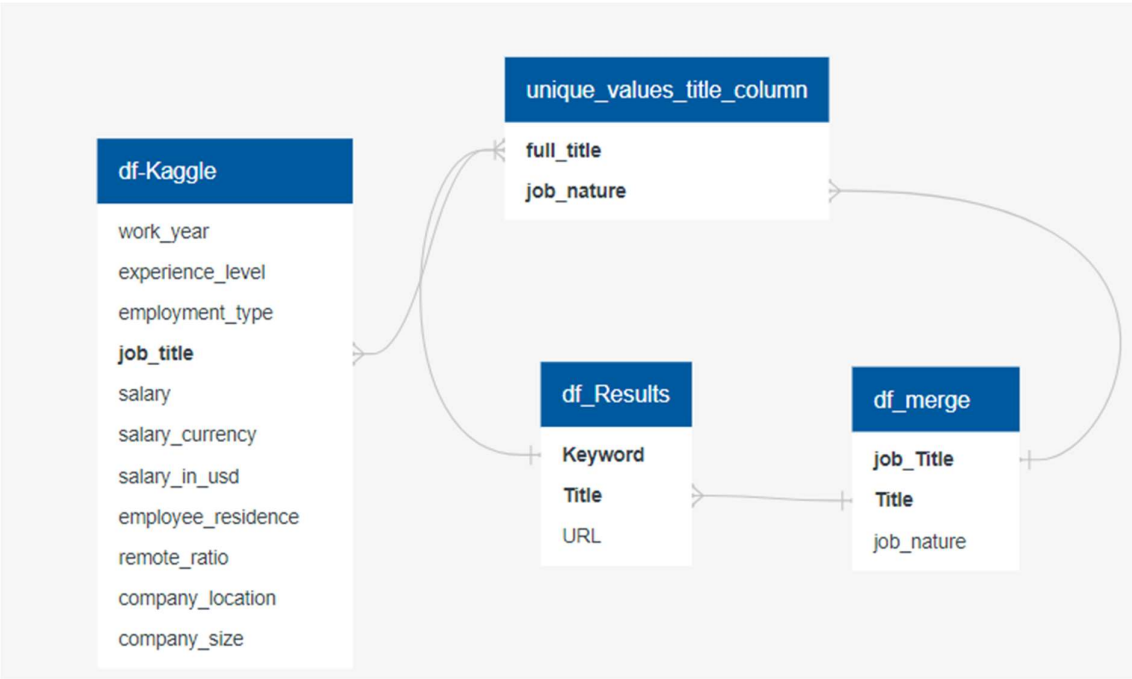

Figure 1: Main data frames (picture created with help of quickdatabasediagrams.com)

3.Data Analysis

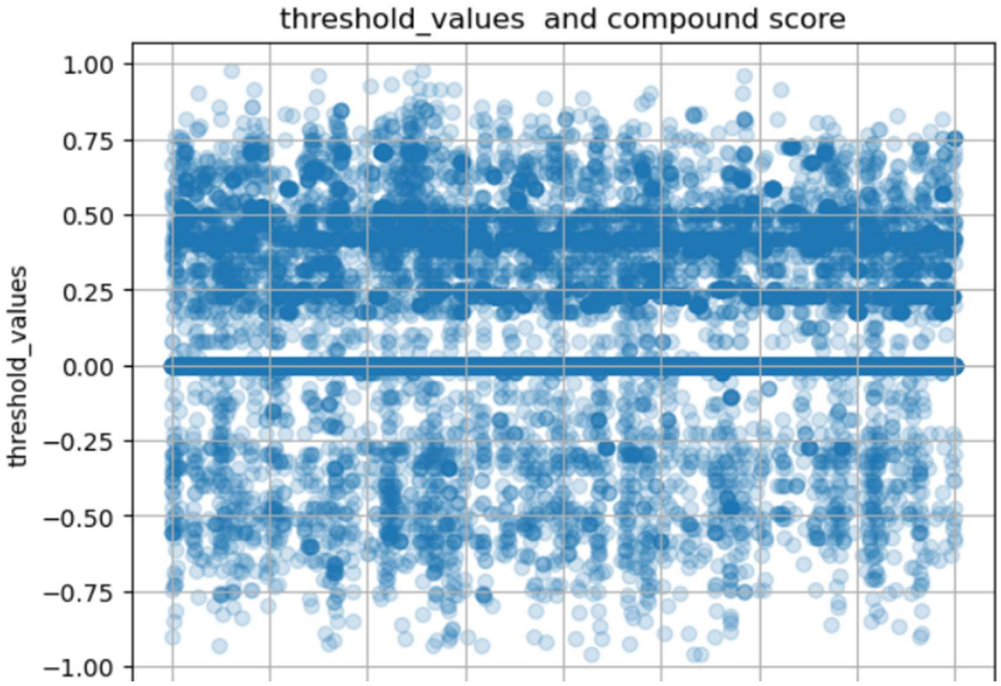

Figure 2: examining the threshold values by plotting compound score.

Firstly, the sentiments of Reddit posts needed to be analysed using VADER (Valence Aware Dictionary and sEntiment Reasoner). VADER is a lexical rule-based analyser designed for analysing short text on social media and other domains (Hutto and Gilbert, 2014; Hutto, no date). According to Hutto and Gilbert (2014), the compound score sums all the lexical ratings normalized between -1 and +1, with +1 denoting the most extreme positive rating and -1 denoting the most extreme negative evaluation. For this study, the polarity scores were used. However, the threshold values are considered as suggestions and not rigid; at the same time, a choice of 0.05 is commonly accepted. For labelling positive, neutral, and negative sentiment based on the compound sentiment score, a scatter plot was created. By examining this scatter plot, a threshold choice of 0.05 was considered, and a new data frame (grouped_SIA) was created, which includes the labels and corresponding value's count percentages. Before further analysis, this newly created data frame was merged with df (Kaggle) to identify the relationship between salary and sentiments.

Label	Threshold	Sentiment	Value Count
0	>=0.05	Neutral	25429
1	<=0.05	Positive	9198
-1	> -0.05) &<0.05	Negative	2229

Figure 3: Value count of each sentiment.

Secondly, the study proceeded to exploratory analysis using correlation to determine the potential relationship between salary and sentiments. The correlation matrix was analysed using two different data frames. The first data frame combined_df, contains numerical features (sentiments and salary) and categorical features (job titles). To identify the correlation of numerical features, particularly salary, standardization was necessary due to its large distribution, which could lead to errors during analysis. To address this, the numerical values were pre-processed using StandardScaler. Standardization, rather than normalization, was chosen based on the assumption that the numerical column values were normally distributed, and to prevent features with higher values from disproportionately impacting the model (*Using StandardScaler() Function to Standardize Python Data | DigitalOcean*, 2022; *What is StandardScaler?*, 2024)Similarly, OneHotEncoder was used to encode the categorical features. This standardization and encoding facilitate the analysis of each job title's relationship with salary. Moreover, correlation was also analysed without standardizing and encoding to get an overall picture of the relationship between sentiments and salary. This approach provides a broader view of the correlation among these variables. As a result, a weak correlation was found in both analyses, and a heatmap was used for visualization.

```
Top 15 variables most correlated with salary:
job_title_Machine Learning Engineer        0.196096
job_title_Research Scientist               0.114158
job_title_Applied Scientist                0.098171
Neutral                                    0.092752
job_title_Research Engineer                0.082487
job_title_ML Engineer                      0.071569
job_title_Data Science Manager             0.056606
job_title_Head of Data                     0.056052
job_title_Machine Learning Scientist       0.054966
job_title_BI Developer                    -0.048167
job_title_Business Intelligence Analyst   -0.056718
job_title_Data Specialist                 -0.062396
job_title_Data Manager                    -0.068599
Positive                                  -0.080483
job_title_Data Analyst                    -0.255916
```

Figure 4: Top 15 variables most correlated with salary.

	Positive	Negative	Neutral	salary
salary	-0.080483	-0.034562	0.092752	1.000000
Neutral	-0.901530	-0.298060	1.000000	0.092752
Negative	-0.144339	1.000000	-0.298060	-0.034562
Positive	1.000000	-0.144339	-0.901530	-0.080483

Figure 5: Variables and correlations.

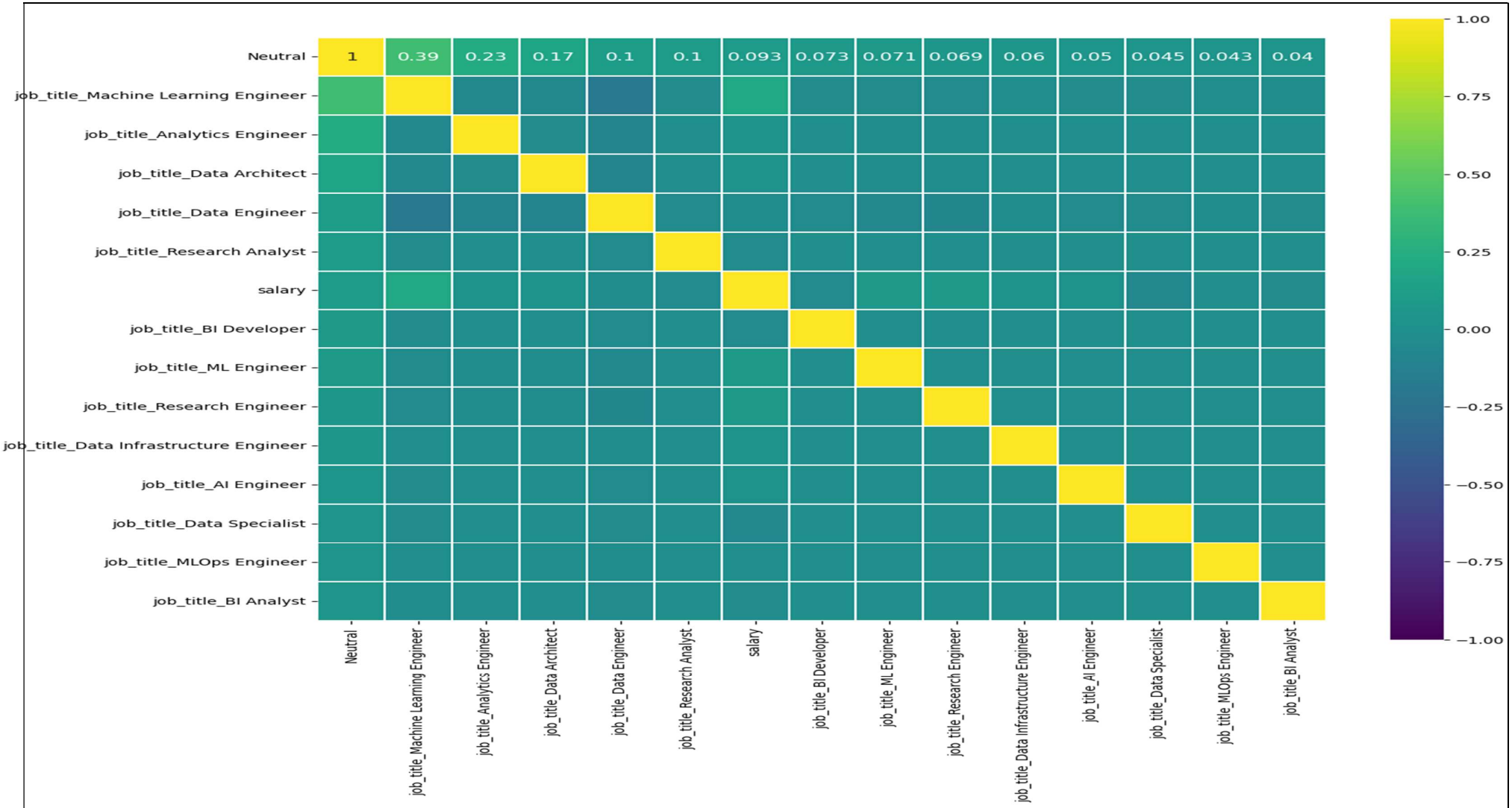

Figure 6: Top 15 variables correlation heatmap of neutral and salary.

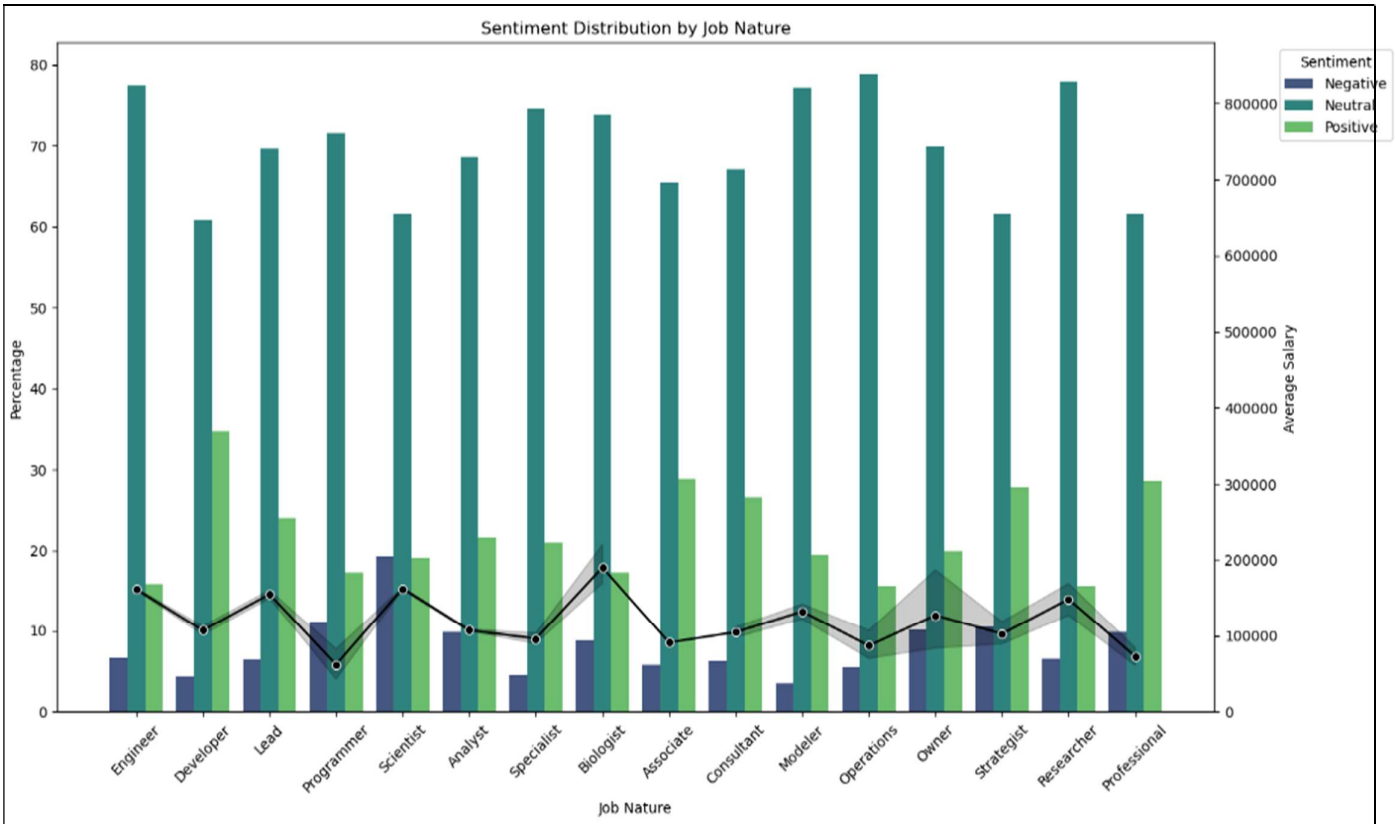

***Figure 7:* grouped bar chart of sentiments by job nature with salary line graph.**

Finally, for better illustration and validation of the correlations, a data frame containing the mean values of sentiments and salary was melted. Visualizing the grouped bar chart provided more clarity, revealing that there are no distinct patterns for classifying data science jobs based on salary and sentiments. Even though regression analysis and clustering does not require the variables to be strongly correlated, it was assumed that regression analysis and other classification techniques, such as K-means clustering, were not performed. Topic modelling was performed as the next level of analysis to identify the underlying topics and their terms in positive, neutral, and negative sentiments. This helps in further understanding how salary as a term is placed with these topics within different sentiments. Latent Dirichlet Allocation (LDA) and Latent Semantic Indexing (LSI) were considered for topic modelling. LDA was chosen due to its probabilistic modelling approach, which identifies groups of topics from the datasets and satisfies the purpose of topic modelling (Pykes, 2023). Data was collected by applying a condition to label from the data frame (df_merge_SIA), then this data was converted into a text file to be used as a corpus. When lambda (λ) is near 0, the positive topic modelling has the term salary, and the negative topic modelling has the term pay, a synonym for salary. However, when λ is close to 1, the terms salary or pay are not found in the top 30 most significant terms. This indicates that salary is not relatively related to the topic model.

4.Conclusions / Recommendations

This project was conducted to examine the relationship between data science domain job salaries and sentiments expressed on Reddit. The correlation matrix is the key analysis tool used in this project. For better understanding and visualization, topic modeling and grouped bar charts were used respectively. The correlation coefficient between the variables and salary was weak, indicating that salary does not have a strong influence on sentiments for data science jobs in Reddit data. Moreover, in all job categories, there is a clear majority of neutral sentiments (more than 60% compound score). In topic modeling, salary did not appear as a top term, further indicating that salary is not a top influential factor. The analysis of positive, negative, and neutral sentiments in the topic modeling datasets (corpus) shows hiring as the top term. This could be due to the nature of job changes within data science jobs. For example, Reddit posts related to data analysts often mention changing careers to data scientist or data engineer roles. Similarly, in negative sentiment topic modeling, the term opposite to hiring appeared as fired. Analyzing the graph, salaries among job natures show variability in average salary, with roles such as engineer, lead, and scientist showing higher salaries. Therefore, there is a relationship between the term hiring and data science jobs. This project also contributes by providing insights into data science professionals who are offering a balanced view of the job market and opportunities.

5.Bibliography

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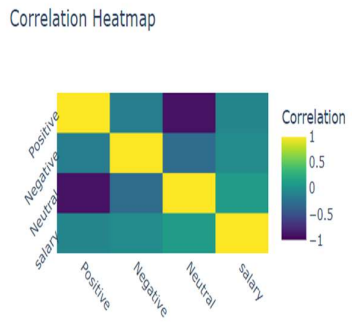
6.Appendix

Dashboard

Correaltion



Veriables Correlation with Salary

index	salary
job_title_Machine Learning Engineer	0.19609635928451616
job_title_Research Scientist	0.11415824704290746
job_title_Applied Scientist	0.09817144360164455
Neutral	0.09275168924117994
job_title_Research Engineer	0.08248729019483078
job_title_ML Engineer	0.07156855885926418
job_title_Data Science Manager	0.056605996630341
job_title_Head of Data	0.05605212816865935
job_title_Machine Learning Scientist	0.0549663989896892

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Grouped Bar Chart